

EMMA SHEDDEN

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EDUCATION

University of Pennsylvania, School of Engineering and Applied Science | Philadelphia, PA

Doctor of Philosophy, *Computer Science*

- GPA: 4.0/4.0, 2 credits completed
- Abel Lukens Stout Fellow

University of Michigan, College of Engineering | Ann Arbor, MI

May 2024, *Summa Cum Laude*

Bachelor of Science in Engineering, *Computer Science* | Minor, *Mathematics*

- GPA: 3.99/4.0, 168 credits completed
- College of Engineering Honors Program
- Select coursework (graduate and honors courses in bold)
 - *Computer science*: Machine Learning, **Reinforcement Learning**, Programming Languages, Compilers, Computation Theory, Data Structures & Algorithms, Computer Security, **Signal Processing**
 - *Mathematics*: **Probability Theory**, Discrete Mathematics, **Extremal Graph Theory**, Theory of Linear Spaces, **Representation Theory**, **Modern Algebra**, Real Analysis, Differential Equations

PUBLICATIONS

Ahmad, H., Endres, M., Newman, K., Santiesteban, P., **Shedden, E.**, & Weimer, W. (2024). Causal Relationships and Programming Outcomes: A Transcranial Magnetic Stimulation Experiment. Accepted to *Proceedings of the 46th International Conference on Software Engineering (ICSE)*. <https://doi.org/10.1145/3597503.3639096>

Chu, S., **Shedden, E.**, Zhang, C., Meira-Góes, R., Moreno, G. A., Garlan, D., & Kang, E. (2023). Runtime Resolution of Feature Interactions through Adaptive Requirement Weakening. In *Proceedings of the 18th Symposium on Software Engineering for Adaptive and Self-Managing Systems*, ser. SEAMS (Vol. 23). <https://doi.org/10.1109/SEAMS59076.2023.00025>

RESEARCH EXPERIENCE

Undergraduate Researcher | **Westley Weimer Lab, University of Michigan**

Sep 2022 - Aug 2024

- Proposed and implemented a blinded data analysis using multilevel linear regression on psychometric data from a 15-participant human study, *accepted to ICSE 2024*
 - Assisted in participant recruitment, transcranial magnetic stimulation (TMS), and fMRI sessions
 - Presented analysis methodology, which was subsequently adapted for use in 2 ongoing research projects
 - Communicated methods and findings in written internal documentation and peer-reviewed publication
 - Made available a complete open-source code artifact for replication of our work
- Designed the coding scheme for a mixed-methods study of online peer tutoring transcripts from nontraditional computer science student interactions during the COVID-19 pandemic
 - Manually annotated chat transcripts (1200+ messages) for use in qualitative and quantitative data analysis
 - Proposed new research directions informed by a comprehensive review of adjacent education literature

Undergraduate Researcher | **LoG(M), University of Michigan**

Jan 2023 - Apr 2023

- Participated in the Lab of Geometry at Michigan under Dr. Hai Le, on a team with 2 other undergraduate students
- Implemented 4 competing algorithms for simulation of standard and fractional Brownian motions, and analyzed their theoretical and observed efficiencies and accuracies
- Delivered tri-weekly PowerPoint progress updates on our group's progress to other members of the program
- Presented our work and final results in a department-wide poster symposium and supporting formal written report

***Undergraduate Researcher* | Eunsuk Kang Lab, Carnegie Mellon University**

Jun 2021 - May 2022

- Participated in the NSF-sponsored Research Experience for Undergraduates in Software Engineering (REU-SE) hosted by the Institute for Software Research (ISR/S3D) as a full-time summer researcher; continued part-time through the 2021-22 academic year
- Collaborated on the design of a novel reconciliation framework to resolve feature conflicts in cyber-physical systems using formal methods and Mixed Integer Linear Programming (MILP), *published in SEAMS 2023*
 - Contributed to the design of a novel grammar to express system requirements and implemented its parser
 - Implemented a case study with three competing quadcopter controllers, and performed testing and data visualization to validate our framework design
 - Migrated and documented an over 10,000-line repository for six competing quadcopter controller frameworks to support future use in testing and analysis
- Collaborated on the design and implementation of a novel autonomous navigation algorithm applying formal methods to continuous Monte Carlo Tree Search
 - Created a suite of new test environments for the training and validation of 2D control algorithms
- Presented research initiatives and findings in a department-wide NSF REU sponsored poster symposium attended by 30 faculty, 50 PhD students, and 40 peers

PROFESSIONAL EXPERIENCE

***Instructional Aide, EECS 203: Discrete Mathematics* | University of Michigan**

Jan 2022 - Apr 2024

- Contributed to course design and development discussions as an IA for 4 semesters
- Planned and delivered weekly one-hour discussions for a class section of 40 undergraduate students
- Collaborated on creation of homework assignments and exams, grading, and day-to-day administration
- Organized weekly two-hour office hours to provide individualized student support for a class of ~1,000 students
- Interviewed 4 candidate instructional aides

***Software Engineer Intern* | Meta Platforms, Inc. (formerly Facebook)**

May 2022 - Aug 2022

- Designed and implemented a generalized service and library enforcing internal access controls on AI assets, scaled to reliably handle over 200 million queries per second using limited cloud resources
- Completed cost-benefit analyses of several internal tools to support server- and client-side caching solutions
- Proposed and implemented design and reliability improvements in multiple other data privacy projects
- Produced quality software with test coverage consistently exceeding my team's benchmark and average
- Communicated my software design and deliverables in three presentations to team members and other engineers

***Supplemental Instruction Leader, Discrete Mathematics* | University of Michigan Ann Arbor**

Feb 2021 - Dec 2021

- Introduced a new IBL-based class structure to increase student engagement and opportunities to collaborate
- Planned and delivered weekly 90-minute supplemental instruction sessions to a class of 15 undergraduate students
- Tutored undergraduate courses including Calculus, Linear Algebra, C++ Programming, and Data Structures
- Organized individual and group office hours to develop effective learning strategies

LEADERSHIP EXPERIENCE

***Drone Subteam Lead* | UM::Autonomy, University of Michigan Ann Arbor**

Sep 2020 - Aug 2022

- Led five team members in developing innovative software solutions to three new challenges in RoboNation's 2022 RoboBoat competition
- Redesigned and implemented an interface for radio communication between an autonomous quadcopter and boat to introduce object-oriented design, expand functionality, and create support for remote testing
- Submitted a technical report on my team's progress and deliverables to the 2021 RoboBoat Competition, including projects in hardware, electrical, and software areas

- Refactored autonomous boat simulation code to be object-oriented, increased simulator functionality, implemented unit tests, and created documentation

SERVICE

Student Volunteer | 37th IEEE/ACM International Conference on Automated Software Engineering *Oct 2022*

- Ran logistics and tech support for hybrid format speaker sessions over 5 days

Student Volunteer | Honors Alternative Spring Break, University of Michigan Ann Arbor *Feb 2022 - Mar 2022*

- Participated in the College of Engineering Honors Alternative Spring Break at Biloxi, Mississippi
- Helped rebuild community gardens and other infrastructure following seasonal hurricane damage over 9 days

SKILLS

Computing Languages: C/C++ (*advanced*); Python, Java, OCaml, R, & Presto (*intermediate*); Golang & HTML/CSS/PHP

Tools: Git, Mercurial, Linux, Visual Studio Code, & MAVLink (*intermediate*); LaTeX & MiniZinc

Languages: Mandarin Chinese (*fluent*), French (*intermediate*)

AWARDS & CERTIFICATIONS

Dean's List and University Honors (*University of Michigan, 2020-23*)

James B. Angell Scholar (*University of Michigan, 2022*)

William J. Branstrom Freshman Prize (*University of Michigan, 2021*)

Advanced Certificate in C++ Programming (*Washtenaw Community College, 2021*)

Michigan Affiliate Award, Aspirations in Computing (*National Center for Women & Information Technology, 2019*)

ACTIVITIES

Ceramist; Artistic Merit award at the NCECA K-12 conference 2018 (*ClayWorkStudio Ann Arbor, 2015-2022*)

Ballet dancer (*Randazzo Dance Company, 2008-2020*)

Traditional Chinese dancer; two Elite Platinum awards at the regional Taoli Cup 2019 (*China Doll Dance Team, 2016-2020*)

Vice President of Operations & Lion Dance Team member (*Formosa Association of Student Cultural Ambassadors, 2016-2020*)